

PROJECT REPORT

2048 Game

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1. Introduction:

2048 is a number combination puzzle based on a grid. The player moves tiles in four directions, and equal adjacent tiles combine into a tile containing their sum. After a successful move, a new tile is placed in an empty position. This project implements the core mechanics in Java while adding a formatted board, colored output, a title header, score display, controls, and styled win, game over, and exit messages.

2. Objectives:

- Develop a functional 2048 puzzle game using Java.
- Implement a 4×4 game board using a two-dimensional array.
- Allow movement in up, down, left, and right directions.
- Combine tiles with equal values.
- Maintain and update the player's score.
- Generate random tiles after successful moves.
- Detect when the 2048 tile has been reached.
- Detect when no further valid move is possible.
- Provide a visually enhanced console interface using ANSI colors.

3. Technologies Used:

Technology / Class	Purpose	Use in Project
Java	Programming language	Implements the complete game logic.
Scanner	User input	Reads W, A, S, D, and Q commands.
Random	Randomization	Selects empty cells and generates 2 or 4 tiles.
ArrayList	Data collection	Stores coordinates of empty board cells.
ANSI Escape Sequences	Terminal styling	Adds colors and formatted output.

4. System Requirements:

- Java Development Kit (JDK).
- Java compatible terminal or IDE terminal.
- Code editor or Java IDE.

- Terminal supporting ANSI escape sequences for enhanced colors.

5. Game Working:

1. Initialize the 4×4 board and score.
2. Generate two initial random tiles.
3. Clear the screen and display the header, board, score, and controls.
4. Read the player's W, A, S, D, or Q command.
5. Perform the corresponding movement function.
6. If the board changes, generate a new random tile.
7. Check whether 2048 has been reached.
8. Check whether another valid move is possible.
9. Continue until the player quits, wins, or reaches game over.

6. Main Functions:

Function	Functionality
main()	Controls initialization, input, movement selection, and game state checks.
printHeader()	Displays the stylized 2048 title and Java Edition label.
printBoard()	Displays the 4×4 board with borders and tile values.
getTileColor()	Returns a terminal color according to tile value.
addRandomTile()	Finds empty cells and inserts a randomly selected 2 or 4.
moveLeft()	Compresses and merges tiles toward the left.
moveRight()	Uses row reversal with left movement.
moveUp()	Uses transposition with left movement.
moveDown()	Uses transposition and reversal with left movement.
reverseRows()	Reverses each board row.
transpose()	Swaps rows and columns.
hasWon()	Checks whether a cell contains 2048.
canMove()	Checks empty cells and adjacent equal tiles.
clearScreen()	Clears the terminal display.
pause()	Waits for Enter after invalid input.

7. User Controls:

Key	Action
W	Move Up
S	Move Down
A	Move Left
D	Move Right
Q	Quit Game

8. Algorithm:

1. Initialize board and score.
2. Place two random tiles.
3. Display the board.
4. Read the player's command.
5. Exit if Q is entered.
6. Perform the selected movement.
7. Generate a new tile after a successful movement.
8. Check for 2048.
9. Check for possible moves.
10. Repeat until the game ends.

9. Future Enhancements:

- Develop a GUI using Java Swing or JavaFX.
- Add actual arrow key controls.
- Add a restart button.
- Store and display the highest score.
- Support different board sizes.
- Add animations and sound effects.
- Add save/load functionality.
- Add a leaderboard system.
- Add difficulty levels.

10. Learning Outcomes:

- Java programming fundamentals.
- Two-dimensional arrays and array manipulation.
- Loops and conditional statements.
- Methods and modular programming.
- Scanner based user input.
- Random number generation.
- ArrayList usage.
- Game state management.
- Basic algorithm design and problem solving.
- ANSI escape sequences for console presentation.

11. Conclusion:

The 2048 Game is a practical application of Java programming concepts in an interactive puzzle game. The project implements a 4×4 board, random tile generation, four direction movement, tile merging, score calculation, win detection, and game-over detection. The enhanced terminal presentation improves the user experience through colors, formatted controls, a title header, and styled game messages. The project also provides a foundation for future development into a graphical Java game.